

SIH PPT DESIGN BRIEF

Maharashtra Skill Intelligence — Slides 2-6

CORE STORY | Industry Demand → Skill Intelligence → Gap Detection → Training Recommendation → Placement Feedback → Continuous Update

Purpose: Production guide for the PPT team. Each slide below gives the message, layout, diagram structure, exact copy to prioritize, and design rules.

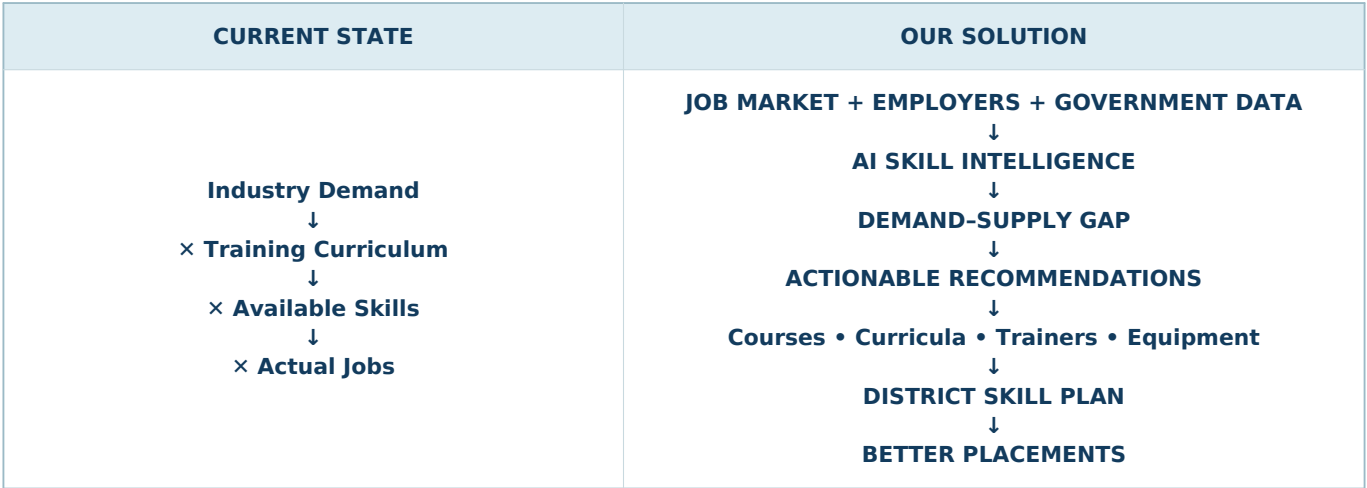
Design direction: Government-tech / data-intelligence aesthetic. Light background, deep blue/teal accents, restrained highlight color, clean cards, thin connectors, strong diagrams, minimal paragraphs.

Positioning: The product is Continuous Skill Alignment + District-Level Decision Intelligence. AI/NLP is the enabling engine—not the product headline.

SLIDE 2 — PROPOSED SOLUTION

Judge objective: In 5-8 seconds, show the current disconnect and the closed-loop intelligence layer you are proposing.

A continuous intelligence layer connecting Industry → Skills → Training → Employment



KEY MODULES — six compact cards; icons/numbers should do most of the visual work.

01 — Labour Market Intelligence Job postings • Employer surveys • Industry trends	02 — Skill Intelligence Skill extraction • Normalization • Emerging skills	03 — Gap Analysis Demand vs Training Supply • District-wise gaps
04 — Training Alignment Course relevance • Curriculum gaps • Obsolete courses	05 — Capacity Planning Seats • Trainers • Equipment • Institutes	06 — Career Intelligence Candidate skill-gap • Career pathways • Local courses

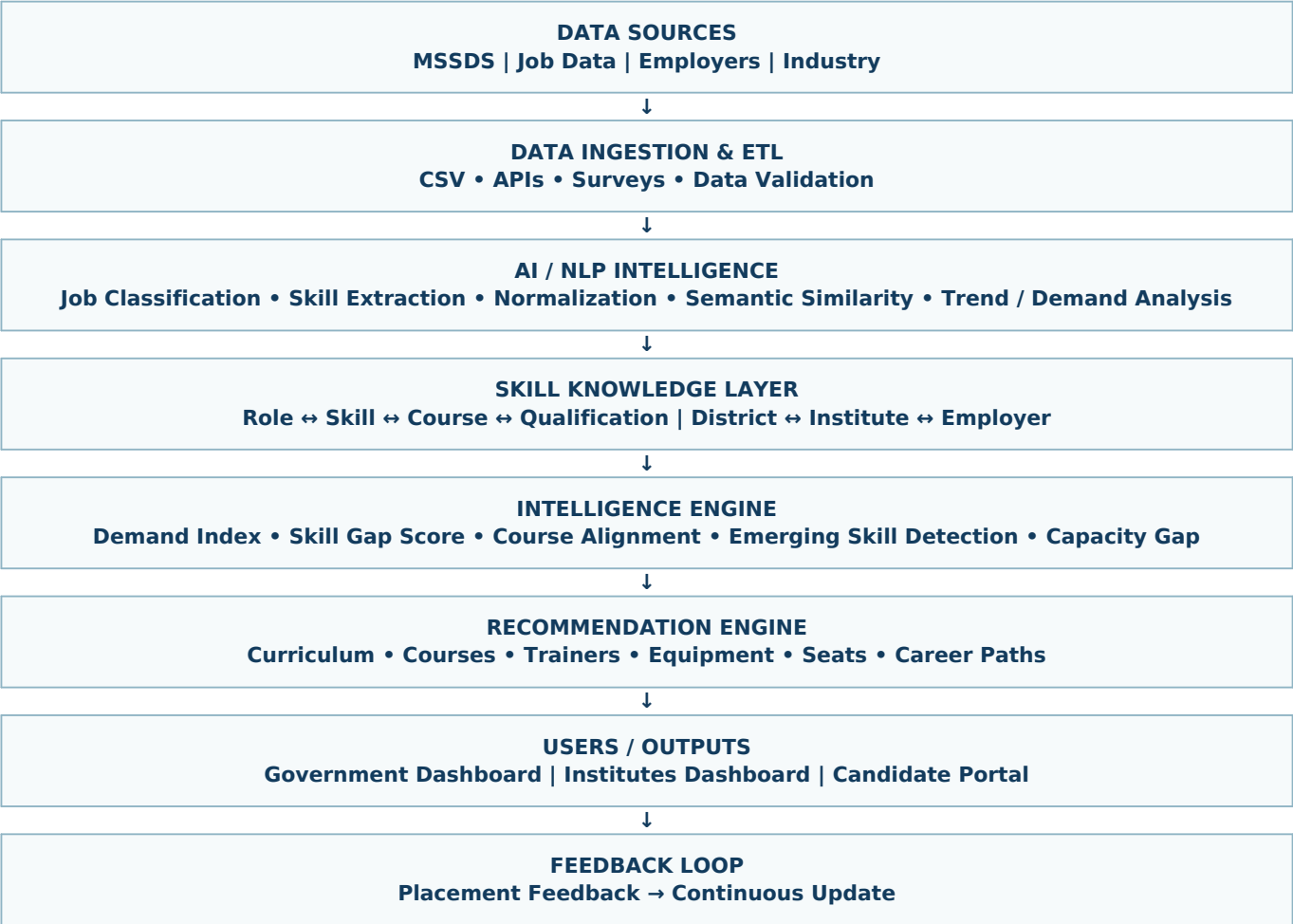
BASELINE → CONTINUOUS UPDATE | FY 2023-24 MSSDS data = baseline • New labour-market signals = continuous updates

Visual rule: The problem/solution split + main pipeline should occupy ~70% of the slide. Avoid a paragraph-heavy layout.

SLIDE 3 — TECHNICAL APPROACH

Judge objective: Prove technical buildability while keeping the architecture readable to a mixed technical/non-technical jury.

ARCHITECTURE — vertical pipeline



TECHNOLOGY STACK — show as a compact right-side stack or footer strip

FRONTEND React / Next.js ECharts / Recharts Leaflet / MapLibre	BACKEND Python FastAPI REST APIs	DATA PostgreSQL pgvector Pandas	AI / ML NLP • Sentence Transformers Embeddings • Scikit-learn LLM for explanation & assistance
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Deployment: Docker • Cloud infrastructure Methodology: Ingest → Clean → Extract → Normalize → Compare → Predict → Recommend → Validate → Learn

TECHNICAL STORY | Data → NLP/AI → Knowledge Layer → Intelligence Scores → Recommendations → Dashboards → Placement Feedback

SLIDE 4 — FEASIBILITY AND VIABILITY

Judge objective: Address data, implementation, AI reliability, privacy, integration and scale risks directly.

WHY FEASIBLE — four cards

□ EXISTING BASELINE DATA MSSDS FY 2023–24 skill-gap / training ecosystem data provides the initial foundation.	□ MATURE TECHNOLOGIES NLP, embeddings, recommendation systems, dashboards and cloud infrastructure are readily implementable.	□ MODULAR ARCHITECTURE New districts, sectors and data sources can be added without redesigning the platform.	□ HUMAN-IN-THE-LOOP Employer validation prevents AI-only policy decisions.
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CHALLENGE → MITIGATION

Challenge	Mitigation
Data freshness	Continuous ingestion + periodic updates
Different skill names	NLP + standardized Skill Taxonomy
Incomplete job data	Combine jobs + employers + government + industry signals
AI inaccuracies	Explainable scoring + human validation
Data privacy	Aggregated/anonymized data + role-based access
Forecast uncertainty	Confidence scores + trend-based predictions
Government integration	API-ready modular architecture

SCALABILITY

PILOT 3-5 DISTRICTS	→	SELECTED SECTORS	→	ALL MAHARASHTRA DISTRICTS	→	MULTI-SECTOR INTELLIGENCE
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USE: “Continuous demand monitoring and evidence-based forecasting.” | AVOID: “real-time accurate prediction of future jobs.”

Bottom line: Start with FY23–24 baseline → validate with current signals → scale continuously.

SLIDE 5 — IMPACT AND BENEFITS

Judge objective: Show ecosystem-wide impact, not just a dashboard feature list.

CENTRAL LOOP — make this the dominant visual



Add a curved return arrow from BETTER TRAINING → INDUSTRY. This loop is the signature visual across Slides 2–5.

STAKEHOLDER BENEFITS

GOVERNMENT Better policy decisions • District-level skill intelligence • Data-driven training investment • Identify oversupplied/obsolete courses • Plan trainers & equipment • Prioritize emerging sectors	TRAINING INSTITUTES Curriculum gap detection • Industry-validated skills • Course modernization • Trainer upskilling • Capacity optimization	EMPLOYERS Job-ready candidates • Reduced skill mismatch • Faster hiring • Employer validation of courses • Industry-training collaboration	CANDIDATES Skill-gap assessment • Local job demand • Personalized learning path • Relevant courses • Improved placement potential
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INTENDED OUTCOMES — clearly label as targets, not achieved results



Impact: Social — better employability & accessible career guidance • Economic — reduced hiring mismatch & optimized public training investment • Institutional — evidence-based curriculum & capacity planning

SLIDE 6 — RESEARCH AND REFERENCES

Judge objective: Establish credibility without a wall of URLs. Use source hierarchy + a clear research-gap statement.

SOURCE GROUPS

GOVERNMENT / PRIMARY Maharashtra State Skill Development Society (MSSDS) Mahaswayam — Skill Development, Employment & Entrepreneurship mahaswayam.gov.in	MSSDS DATA / ECOSYSTEM FY 2023-24 Skill Gap Analysis Report Training Centres • Courses • Job Roles • District information kaushalya.mahaswayam.gov.in	ECONOMIC / SKILL STANDARDS Directorate of Economics & Statistics, Maharashtra — mahades.maharashtra.gov.in NSDC — NOS / skill framework resources NSQF — qualification & competency alignment	RESEARCH AREAS Labour Market Intelligence • NLP-based Skill Extraction • Skill Taxonomy / Knowledge Graphs • Human-in-the-loop AI
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RESEARCH GAP

Existing skill-gap studies provide valuable snapshots, but labour-market requirements continuously evolve. Our approach creates a continuous feedback loop connecting labour demand, skills, training supply, curriculum, capacity and placement outcomes.

QR CODE / PROTOTYPE ZONE

Prototype & Technical Documentation Place the GitHub / demo QR here by submission time.	 QR
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JUDGE TAKEAWAY | “We continuously compare what Maharashtra trains with what Maharashtra’s industries need—and convert the gap into actionable training decisions.”

DESIGN SYSTEM — APPLY ACROSS ALL 6 SLIDES

Visual language: Government-tech / data-intelligence. Light background, deep navy/blue headers, teal secondary elements, one restrained highlight color, rounded cards, thin connectors, simple line icons. Avoid decorative stock illustrations.

Hierarchy: One message headline → one dominant visual → supporting cards/evidence → one bottom takeaway. Do not give equal visual weight to every bullet.

Diagram rules: Arrows = flow. × = current-state break. Loop arrow = feedback. Keep labels short enough for projection.

Typography: Large bold titles; small uppercase section labels; medium/bold diagram labels; short noun phrases instead of long sentences.

Do not: Repeat “AI-powered” everywhere. Do not claim perfect/real-time future-job prediction. Do not show targets as achieved results. Do not invent accuracy percentages.

PRODUCT DIFFERENTIATOR | Continuous Skill Alignment + District-Level Decision Intelligence

Final test: A judge seeing Slides 1-5 for one minute should understand: what is broken → what you built → how it works → why it is feasible → what changes.